

Full length article

Probiotics ameliorate polyethylene microplastics-induced liver injury by inhibition of oxidative stress in Nile tilapia (*Oreochromis niloticus*)

Rui Dong, Changlei Zhou, Shuyue Wang, Yisha Yan, Quan Jiang^{*}

Key Laboratory of Bio-resources and Eco-environment of Ministry of Education, College of Life Sciences, Sichuan University, Chengdu, PR China

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ABSTRACT

Microplastic particles (MPs) are environmental pollutants that can cause varying levels of aquatic toxicity. Probiotics have been shown to reduce the negative effects of toxic substances. However, the protective effect of probiotics against the adverse effects of MPs has yet to be reported. The current study sought to determine the effects of the commercial probiotic AquaStar® Growout on polystyrene (PS)-MPs-mediated hepatic oxidative stress in Nile tilapia (*Oreochromis niloticus*). Fishes were assigned into four groups: the first group was the control, the second group was exposed to 1 mg/L of 0.5 µm PS-MPs, and the third and fourth groups were exposed to 1 mg/L of 0.5 µm PS-MPs and pre-fed with probiotics at levels of 3 g/kg and 6 g/kg diet, respectively. At the end of the experiment, probiotics administration reversed liver damage caused by the PS-MPs, reducing serum levels of malondialdehyde, aspartate aminotransferase, and alanine aminotransferase, and increasing the total antioxidant capacity. Furthermore, probiotics alleviated PS-MPs-induced oxidative stress by restoring antioxidant enzyme activities (superoxide dismutase, catalase, glutathione S-transferase, and glutathione peroxidase) and reducing oxidized glutathione and enhancing the redox state. Besides, probiotics supplementation decreased the transcriptional level of C-reactive protein and tumor necrosis factor-α following PS-MPs exposure. Furthermore, probiotics counteracted PS-MPs-associated reactive oxygen species production and mitogen-activated protein kinases (MAPKs) phosphorylation status. These findings suggested that probiotics could decrease liver damage caused by PS-MPs through their antioxidant properties and modulation of MAPK signaling pathways.

1. Introduction

Plastic litter in natural environments is a global issue, accounting for up to 50–80% of all waste in oceans and beaches worldwide [1]. There is a growing concern regarding the presence of microplastic particles (MPs, with a diameter <5 mm) in aquatic environments derived from primary and secondary plastic litter [2]. The most commonly manufactured plastics, such as polystyrene (PS), polyethylene (PE), polypropylene (PP), and polyethylene terephthalate (PET), have a carbon backbone that is resistant to hydrolytic and enzymatic degradation. Since PS-based plastic was one of the main high-production plastic polymers representing 90% of total plastic demand [3], the presence of PS-MPs contamination has been widely reported in marine [4], estuarine [5], and freshwater environments [6]. PS-MPs can cross the intestinal barrier [7] and translocate to multiple fish organs, including the liver, brain, and heart [7–9]. Several lines of evidence strongly implicated the roles of PS-MPs in the induction of oxidative stress and organ damage in fish models [10–13]. In addition, studies demonstrated that

oxidative stress caused by PS-MPs exposure activated the mitogen-activated protein kinases (MAPKs) signaling pathways in marine species such as the monogonont rotifer (*Brachionus koreanus*) [14], the Chinese mitten crab (*Eriocheir sinensis*) [15], and the copepod (*Paracyclops nana*) [16], as well as the invertebrate organism of *Daphnia magna* [17]. However, very little is known about the direct effects of PS-MPs on MAPK signaling pathways in fish.

Probiotics are live microorganisms that provide health benefits to the host [18]. Consumption of probiotics alone or in food demonstrates that probiotics can exhibit antioxidant activity and reduce oxidative damage [19]. The antioxidant effects of probiotics include oxidant compound scavenging, metal ion chelation, increased antioxidant activity, activation of the antioxidant signaling pathway, and regulation of reactive oxygen species (ROS)-producing enzymes [19,20]. The antioxidant response of tilapia was shown to improve after exposure to various probiotics, including *Aspergillus oryzae* [21], *Bacillus subtilis* [22], *Bacillus licheniformis* [23], and *Bacillus species* and yeast [24]. AquaStar® Growout (Biomim Holding GmbH, Austria) is a lyophilized probiotic mixture containing *Bacillus subtilis*, *Enterococcus faecium*, *Lactobacillus*

^{*} Corresponding author.

E-mail address: jiangqua@gmail.com (Q. Jiang).

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Abbreviations

PS-MPs	polystyrene microplastic particles
AST	aspartate transaminase
ALT	alanine transaminase
MDA	malondialdehyde
T-AOC	total antioxidant capacity
SOD	superoxide dismutase
CAT	catalase
GST	glutathione S-transferase
GR	glutamic reductase
GPx	glutathione peroxidase
GSH	reduced glutathione
GSSG	oxidized glutathione
CRP	C-reactive protein
TNF α	tumor necrosis factor- α ;
IL-1 β	interleukin-1 β
ROS	reactive oxygen species
MAPK	Mitogen-activated Protein Kinase
Erk	Extracellularly Regulated Kinase
JNK	c-Jun N-terminal kinases

reuteri, and *Pediococcus acidilactici*, which could improve antioxidative capacity and immunity in Nile tilapia (*Oreochromis niloticus*) [25,26] and rainbow trout (*Oncorhynchus mykiss*) [27]. Besides, AquaStar® Growout application could enhance growth performance and immune parameters in whiteleg shrimp (*Litopenaeus vannamei*) [28]. However, the potential protection of probiotics against MPs-induced oxidative stress remains unknown in vertebrates.

Therefore, the purpose of the present study was to test the hypothesis that probiotic supplementation may mitigate PS-MPs-induced oxidative stress. The Nile tilapia (*Oreochromis niloticus*) is one of the most widely farmed fish species in the world. Multiple field studies have reported MPs contamination in tilapia as a result of rapid industrialization and urbanization along the river [29,30]. MPs have been shown in laboratory studies to cause oxidative stress [31], neurotoxicity [32], reproductive impairment [33], anemia and perturbations in hemato-biochemical parameters [34], and dysbiosis of the gut microbiota [35] in tilapia. This study aimed to assess the potential protective effects of dietary supplementation with the commercial multi-strain probiotic AquaStar® Growout on PS-MPs-induced oxidative stress by evaluating hepatic histological changes, serum biochemical parameters, enzyme and non-enzymatic antioxidant levels, and gene expression of inflammation markers in Nile tilapia. Also, probiotics contributed to the defense mechanism against PS-MPs-induced oxidative stress by examining ROS production and the phosphorylation status of various MAPK signaling proteins.

2. Materials and methods

2.1. Reagents and antibodies

PS-MPs beads with diameters of 0.5 μm (density: 1.05 g cm⁻³) were purchased from Yuan Biotech (Shanghai, China). The stock was supplied as dispersions in deionized water with a mass concentration of 10 mg/mL (approximately 7.48×10^9 particles/mL). PS-MPs were added to each aquarium at a concentration of 1 mg/L, which was within the environmentally relevant freshwater levels [36]. The morphology and size of PS-MPs were examined and photographed by using scanning electron microscopy (Zeiss, Sigma-300, Germany) (Fig. S1). N-acetyl-L-cysteine (NAC), a common antioxidant to inhibit oxidative stress, was obtained from Sigma (St Louis, MO). Antibodies for phosphorylated p38 MAPK (p-p38 MAPK), anti-p38 MAPK, phosphorylated extracellular

signal-regulated kinase (p-ERK1/2), anti-ERK1/2, phosphorylated c-Jun N-terminal kinase (p-JNK), anti-JNK, and anti- β -actin polyclonal antibody were purchased from Cell Signaling Technology (Danvers, MA, USA). As secondary antibodies, HRP-conjugated goat-anti-rabbit-IgG antibody from Santa Cruz Biotechnologies (Santa Cruz, CA) was used.

2.2. Culture of tilapia

Healthy Nile tilapia (*Oreochromis niloticus*) were randomly stocked in multiple 24-L fiberglass tanks (initial bodyweight: 4.34 ± 0.22 g) and maintained in freshwater aquaria at 28 °C under 12 h dark/12 h light photoperiod. Fish were fed on a commercial diet (40% protein, 12% fat, 2% fiber, 8.5% moisture, 8% ash, Tongwei, China) for two weeks to adjust to the experimental conditions. Before the feeding experiment, a visual examination of the fish's health was conducted by observing their movements, appearance, and signs of infectious diseases. The water quality was monitored daily and maintained at pH = 6.5 ± 0.5 (adjusted with NaHCO₃ as necessary) and dissolved oxygen >7 mg/L. Fish maintenance and experimental procedures were approved by the Committee on Animal Research and Ethics of Sichuan University.

2.3. Preparation of experimental diets

The commercial diet was supplemented with multi-species probiotics AquaStar® Growout containing *Bacillus subtilis*, *Enterococcus faecium*, *Lactobacillus reuteri*, and *Pediococcus acidilactici* (Biomim Holding GmbH, Austria). The proportions of the bacterial species are approximately equal at a total cell count of 1×10^9 colony forming units (CFU)/g. Two dosages of AquaStar® Growout were top-coated on the feed at 3 g/kg (4.3×10^8 CFU/g of diet) and 6 g/kg (7.5×10^8 CFU/g of diet). These dosages were determined based on a previous report [37]. To incorporate probiotics into the diet, 3 g or 6 g of probiotics for each kg of feed were carefully mixed with 10 mL of soybean oil under constant stirring. Only the same amount of soybean oil was added to the control diet. The diets were stored under refrigeration at 4 °C. Fresh diets were prepared every two weeks to prevent probiotic viability from diminishing.

2.4. Experimental design and samplings

The experimental fishes were randomly assigned into four groups: the control group where the fish were continuously fed on the basal diet (without probiotics), the second group where the fish were continuously fed on the basal diet (without probiotics), and then exposed to PS-MP-polluted water (1 mg/L of 0.5 μm PS-MPs) for extra two weeks, and two test groups where the fish were continuous feeding of the basal diet supplemented with AquaStar® Growout probiotics (3 g/kg and 6 g/kg, respectively), and then PS-MP-polluted water for extra two weeks. The specific experimental protocol is shown in Fig. 1A. For ROS assays and Western blotting studies, the fish were separated into four treatment groups: (1) Control group, in which fish were continuous feeding a basal diet and then injected daily PBS with 100 μL (i.p.) for extra two weeks, (2) PS-MPs group, in which fish were continuous feeding basal diet and then injected daily PBS with 100 μL (i.p.) and exposed to PS-MP-polluted water (1 mg/L of 0.5 μm MP) for extra two weeks, (3) PS-MPs + probiotics (P), in which fish were continuous feeding of the basal diet supplemented with AquaStar® Growout probiotics (6 g/kg), and then fish were injected PBS daily with 100 μL (i.p.) and exposed to PS-MP-polluted water (1 mg/L of 0.5 μm PS-MPs) for extra two weeks, and (4) PS-MPs + NAC, in which fish were continuous feeding basal diet and then daily injected 100 mg/kg/d NAC (i.p.) and exposed to PS-MP-polluted water (1 mg/L of 0.5 μm PS-MPs) for extra two weeks. The specific experimental protocol is described in Fig. 1B. Diets were administered three times a day (0900, 1230, and 1800 h) to achieve satiety. During the experiment, the test solutions were replaced daily, and the tanks were continuously aerated to keep the PS-MPs dispersed in water throughout the exposure periods. Fish were fasted for 24 h and

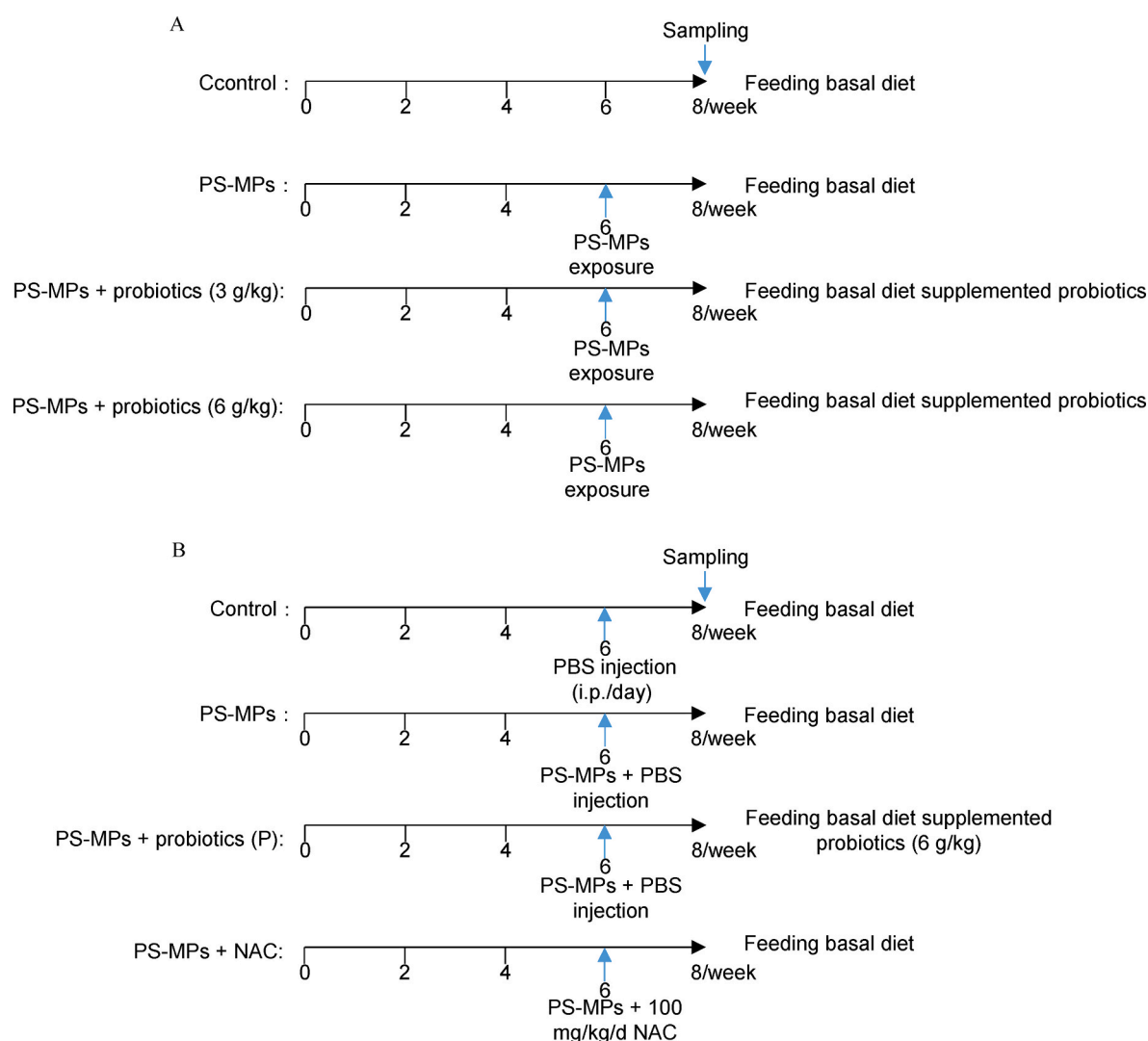


Fig. 1. Experimental design and animal exposure. (A) Experimental protocol for histological analysis and determination of the selected oxidative stress parameters, and (B) Experimental protocol for ROS production and Western blotting.

then sacrificed with an overdose of tricaine methanesulfonate (MS 222; 250 mg/L) prior to sampling. The serum and liver samples were collected and stored at -80°C to analyze the oxidative stress and inflammation indicators.

2.5. Culture-based analysis of intestine probiotics

To detect the presence of probiotic bacteria, the fecal matter from the mid-intestine was isolated [25] and pooled by the same tank ($n = 3$) for culture-dependent analyses at the end of eight weeks of experimental feeding. The samples were serially diluted in PBS, and 20 μL was spotted on duplicate MRS agar plates, Slanetz and Bartley plates, and Bacillus selective plates using the Miles and Misra method [38]. The total populations of heterotrophic aerobic bacteria were determined using tryptone soya agar. After 48 h of incubation at 28°C , colonies on statistically viable plates are expressed as base-10 logarithm CFU/g of digested material. The 16 S rRNA gene sequence analysis was used to confirm representative subsets of the presumptive probiotics [39].

2.6. Histological analysis

After eight weeks of experimental feeding, the liver tissues were fixed in 10% formalin and embedded in paraffin. The embedded

material was sectioned to a thickness of 5 μm . Afterward, samples were stained with hematoxylin-eosin (H&E) dyes. Then, they were examined using an upright microscope (Ni-U, Nikon, Japan).

2.7. Measurements of serum oxidative stress indicators

Blood was collected from the caudal vein using heparinized syringes. Following centrifugation at 3000 g for 10 min at 4°C , plasma was obtained and stored at -80°C for further analysis. The alanine aminotransferase (ALT) and aspartate aminotransferase (AST) levels in serum samples were assessed using a chemistry analyzer (AU480; Beckman Coulter, CA, USA). The malondialdehyde (MDA) and the total antioxidant capacity (T-AOC) were determined using commercial kits obtained from the Nanjing Jiancheng Bioengineering Institute (Nanjing, China). MDA was determined by thiobarbituric acid reaction at 532 nm [40]. The MDA content was expressed as $\mu\text{mol/L}$. T-AOC antioxidant capacity was determined by the ferric reducing/antioxidant power method at 593 nm [41], and the result was expressed as mmol/L .

2.8. Measurements of enzymatic and non-enzymatic antioxidant compounds in liver samples

The liver tissues were lysed using TissueLyser II (Qiagen, Hilden,

Germany), and then centrifuged at 13,000 rpm at 4 °C for 15 min. The supernatants were separated into aliquots and stored at –80 °C until antioxidant enzymes. The superoxide dismutase (SOD), catalase (CAT), glutathione S-transferase (GST), glutathione peroxidase (GPx), glutamic reductase (GR), reduced glutathione (GSH), and oxidized glutathione (GSSG) were performed using commercial kits obtained from the Nanjing Jiancheng Bioengineering Institute (Nanjing, China). The procedures were carried out in accordance with the manufacturer's instructions. Briefly, SOD activity was measured by referring to the method of McCord and Fridovich [42] at 450 nm. One unit (U) of SOD activity represented the amount of enzyme that caused a 50% inhibition of cytochrome c. The SOD activity was expressed as U/mg protein. CAT activity was assayed by the method of Göth [43] at 405 nm. One U of CAT activity was equivalent to the quantity of enzyme breaking down one mM of hydrogen peroxide per min. The CAT activity was expressed as U/mg protein. GST activity was defined by catalyzing the reduction of GSH binding to substrate 1-chloro-2,4-dinitrobenzene [44]. One enzyme activity U is defined as a reduction in GSH concentration by one μ mol per min. The GST activity was represented as U/mg protein. GPx activity was monitored by following the decrease of GSH at 412 nm using hydrogen peroxide (H₂O₂) as substrate [45]. One U of GPx activity is the amount of enzyme that depleted one mmol of GSH per min. The GPx activity was represented as U/mg protein. GR activity was monitored by measuring the decrease of nicotinamide adenine dinucleotide phosphate at 340 nm [46]. The GR content was represented as U/g protein. The concentrations of GSH and GSSG were determined by colorimetric analysis at 405 nm [47]. GSH generated from the excess of GSSG reduced 5,5'-dithiobis (2-nitrobenzoic acid). The contents of GSH and GSSG were represented as μ mol/g protein.

2.9. Measurement of C-reactive protein (CRP), tumor necrosis factor- α (TNF α), and interleukin-1 β (IL-1 β) gene expression levels by real-time PCR (qPCR)

Total RNA from the livers was isolated with TRIzol reagent (Invitrogen) followed by chloroform/isopropanol extraction. RNA was quantified using NanoDrop (ND-1000) spectrophotometer (Thermo Scientific). After digestion with DNase I to remove genomic DNA contamination, total RNA (1 μ g) was reverse transcribed using iScriptTM cDNA synthesis kit (Bio-Rad). qPCR was performed on a Mastercycler[®] ep realplex qPCR system (Eppendorf) and analyzed using Realplex 2.2 (Eppendorf). qPCR reaction was tested with an SYBR Select Master Mix kit (Invitrogen) using the primer pairs for tilapia CRP (forward primer: 5' CCAGGGGCAAGGCGAATAAAG 3' and reverse primer: 5' GTTCTGGATCTTGAACGGG 3'), TNF α (forward primer: 5' ATGTGA-GAGCAGCCATTCAT 3' and reverse primer: 5' ACAAAGTA-GAGGCCATCTCG 3') and IL-1 β (forward primer: 5' CAGTGAAGACCGCAAAGTGC 3' and reverse primer: 5' CGGTCTCTGACACACTTCC 3'). After an initial incubation at 95 °C for 2 min, 35 cycles of the following temperature conditions were used: 94 °C for 30 s, 55 °C for 30 s, and 30 s at 72 °C with fluorescence signal detection at the end of each cycle, followed by melting curve of the amplified products obtained by ramped increase of the temperature from 55 to 95 °C to determine the presence of single amplification product per reaction. In this case, β -actin (forward primer: 5' GTGATGGTGGGTATGGGT 3' and reverse primer: 5' GGCAACTCT-CAGCTCGTT 3') was used as an internal control for normalizing genes of interest, and transcript levels were analyzed using the 2^{– $\Delta\Delta$ CT} method.

2.10. Reactive oxygen species

The liver tissues were lysed and then centrifuged at 13,000 rpm at 4 °C for 15 min. The supernatants were combined with 2',7'-dichloro-fluorescein (DCF) yellow diacetate probe and incubated in the dark for fluorescence labeling in a 96-well plate. The intensity of fluorescence was measured at excitation wavelengths of 485 nm and emission

wavelengths of 520 nm. According to the kit manufacturer's recommendations (Nanjing Jiancheng Bioengineering Institute, China), and results were expressed as U DCF per mg of protein.

2.11. Western blot

Western blot was performed as described previously [48]. Briefly, tilapia liver samples with the indicated treatments were lysed with RIPA buffer (Abcam, Cambridge, UK) containing protease inhibitors (Roche Diagnostics, Mannheim, Germany). The lysate was centrifuged at 13,000 g for 10 min at 4 °C to clear the cellular debris. Proteins were quantified using the Bradford protein assay kit (Bio-Rad, Hercules, CA). The same amount of proteins was resolved on 10% SDS-PAGE. After that, they were electrophoretically transferred to a polyvinylidene difluoride membrane (Amersham Pharmacia Biotech, Uppsala, Sweden). Blots were probed with the indicated antibodies. These antisera have been used previously in tilapia [48]. Immunoreactive proteins were detected using ECL plus chemiluminescence reagent (Amersham Pharmacia Biotech) following the manufacturer's instructions.

2.12. Statistical analysis

Data are expressed as the mean $\pm$ standard error of the mean (SEM). Data were analyzed by multiple comparison tests using one-way analysis of variance (ANOVA) followed by post-hoc Bonferroni's test. The *p*-value less than 0.05 was considered statistically significant. Unless otherwise indicated, all experiments were done a minimum of three times, with one representative experiment shown. All graphs and statistical analyses were done using GraphPad Prism Software, Inc (San Diego, CA).

3. Results

3.1. Culturable intestinal microbiota

To determine the effect of the AquaStar[®] Growout treatments on intestinal allochthonous probiotic population, culture-based methods were employed for bacterial analyses. As shown in Table 1, no significant differences were detected in the total viable cell counts (TVC) between the treatments, with allochthonous levels fluctuating around a log 6 CFU/g level for each treatment at the end of the experiment. However, lactic acid bacteria (LAB) and *Bacillus* spp. significantly decreased in the PS-MPs-exposed group compared to the control group. In contrast, LAB and *Bacillus* spp. populations were significantly higher in co-treatments of PS-MPs + pre-feeding probiotics (3 g/kg) group and PS-MPs + pre-feeding probiotics (6 g/kg) group when compared to the control and PS-MPs group. In addition, the PS-MPs + pre-feeding probiotics (6 g/kg) group contained the highest levels of *enterococci* (log 4.72 CFU), which were significantly higher than those found in other treatments.

Table 1

Allochthonous TVC, LAB, *Bacillus* spp., and *enterococci* (log CFU/g) in the intestinal tract of tilapia after feeding on experimental diets.

Species	Control	PS-MPs	PS-MPs + Probiotic (3 g/kg)	PS-MPs + Probiotic (6 g/kg)
TVC	6.07 $\pm$ 0.39a	5.94 $\pm$ 0.55a	6.18 $\pm$ 0.78a	6.23 $\pm$ 0.34a
LAB	2.26 $\pm$ 0.35a	0.74 $\pm$ 0.45b	4.57 $\pm$ 0.71c	5.76 $\pm$ 0.47c
<i>Bacillus</i> spp.	2.08 $\pm$ 0.34a	0.61 $\pm$ 0.33b	4.18 $\pm$ 0.44c	5.53 $\pm$ 0.61c
<i>Enterococci</i>	n.da	n.da	2.25 $\pm$ 0.49b	4.72 $\pm$ 0.77c

n.d = not detected.

Different superscripts indicate a significant difference (*p* < 0.05).

3.2. Liver histological analysis

To determine whether tilapia with pre-feeding probiotics protects the liver from oxidative damage caused by PS-MPs, liver tissues were stained with H&E. In the control group, the majority of hepatocytes with polygonal shapes had round or oval nuclei, and their boundaries were well-defined (Fig. 2A). However, histological changes were observed in the liver of tilapia exposed for two weeks to 1 mg/L of 0.5 μ m PS-MPs. These signs included vacuolation in hepatocytes, degeneration of nuclei, erythrocyte infiltration into blood sinusoids, damage to hepatocyte palisade arrangements, and compromised cell boundaries (Fig. 2B). In the PS-MPs + probiotics (3 g/kg) group, probiotic administration attenuated the adverse effects caused by PS-MPs exposure, resulting in slight vacuolation of hepatocytes (Fig. 2C). In contrast, no histological abnormalities were observed in liver tissues of the PS-MPs + probiotics (6 g/kg) group (Fig. 2D).

3.3. Effects of probiotics on serum antioxidative status in PS-MPs-treated tilapia

To access the hepatic damage caused by oxidative stress, ALT and AST levels were measured in the serum of tilapia administered with probiotics and PS-MPs. As shown in Fig. 3A and B, compared to the control, there was a significant rise in ALT and AST levels after two weeks PS-MPs exposure in tilapia. However, pretreatment with

probiotics (3 and 6 g/kg) markedly alleviated ALT and AST levels in the serum of PS-MPs-treated tilapia. The level of MDA, an important marker of lipid peroxidation, was measured in serum after exposure to PS-MPs with pre-feeding of the probiotic. As depicted in Fig. 3C, the PS-MPs-exposed group increased the concentration of serum MDA. Conversely, in the PS-MPs-exposed group pretreated with probiotics, the MDA level was similar to that in the normal control group or significantly lower than in those treated with PS-MPs only group. T-AOC of plasma is an index of the redox buffer capacity. In our case, the T-AOC level was inhibited in the liver after exposure to PS-MPs compared with the control, whereas the PS-MPs-exposed liver pretreated with probiotics restored the T-AOC levels similar to that of the control (Fig. 3D).

3.4. Probiotics restored PS-MPs-induced antioxidant enzyme activities and non-enzymatic antioxidant levels in tilapia liver

To test if probiotics can protect PS-MPs-exposed liver from oxidative damage, the activities of antioxidant enzymes are measured. Compared with control, the activities of SOD (Fig. 4A), CAT (Fig. 4B), GST (Fig. 4C), and GPx (Fig. 4D) were notably increased without affecting GR activity (Fig. 4E) after two weeks of PS-MPs exposure. However, the elevations of SOD, CAT, GST, and GPx levels were inhibited in pre-treatment with probiotics at doses of 3 and 6 g/kg compared to the PS-MPs-treated group. Moreover, the co-treatment did not cause a marked change in GR activity. Changes in the contents of non-enzymatic

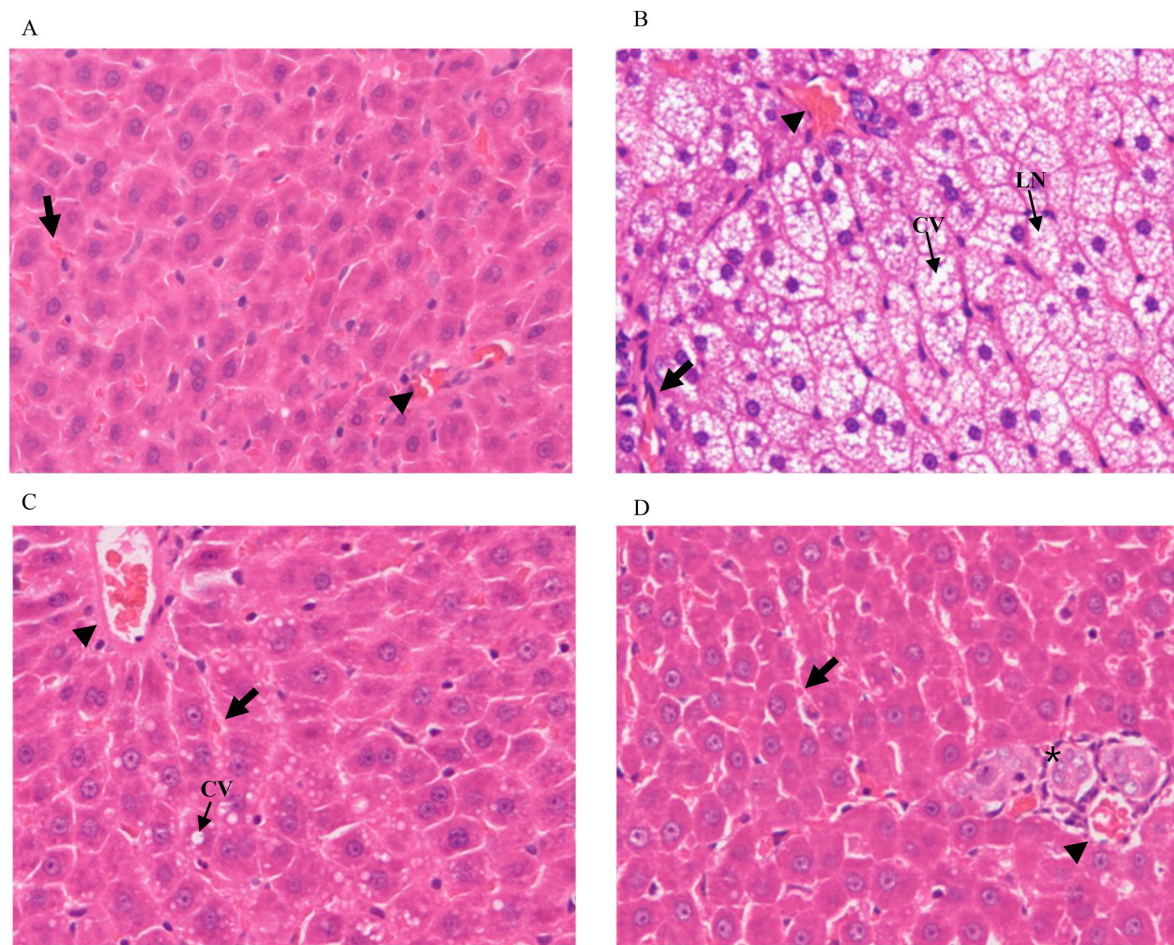


Fig. 2. Representative histological section analysis of protective effects of probiotics on PS-MPs-induced liver injury stained by hematoxylin-eosin. (A) Control: no morphological changes. (B) PS-MPs exposure group: showing degenerated nuclei, vacuolation, and erythrocyte infiltration into blood sinusoids. (C) PS-MPs + probiotics (3 g/kg) group: showing slight vacuolation in hepatocytes. (D) PS-MPs + probiotics (6 g/kg) group: no morphological changes. The slides were examined under the light microscope at a magnification of $\times 200$. Arrowhead, hepatic vessel; Asterisk, bile duct; Arrow, sinusoid; "CV", represents cytoplasmic vacuolation and "LN", represents lysis of nucleolus.

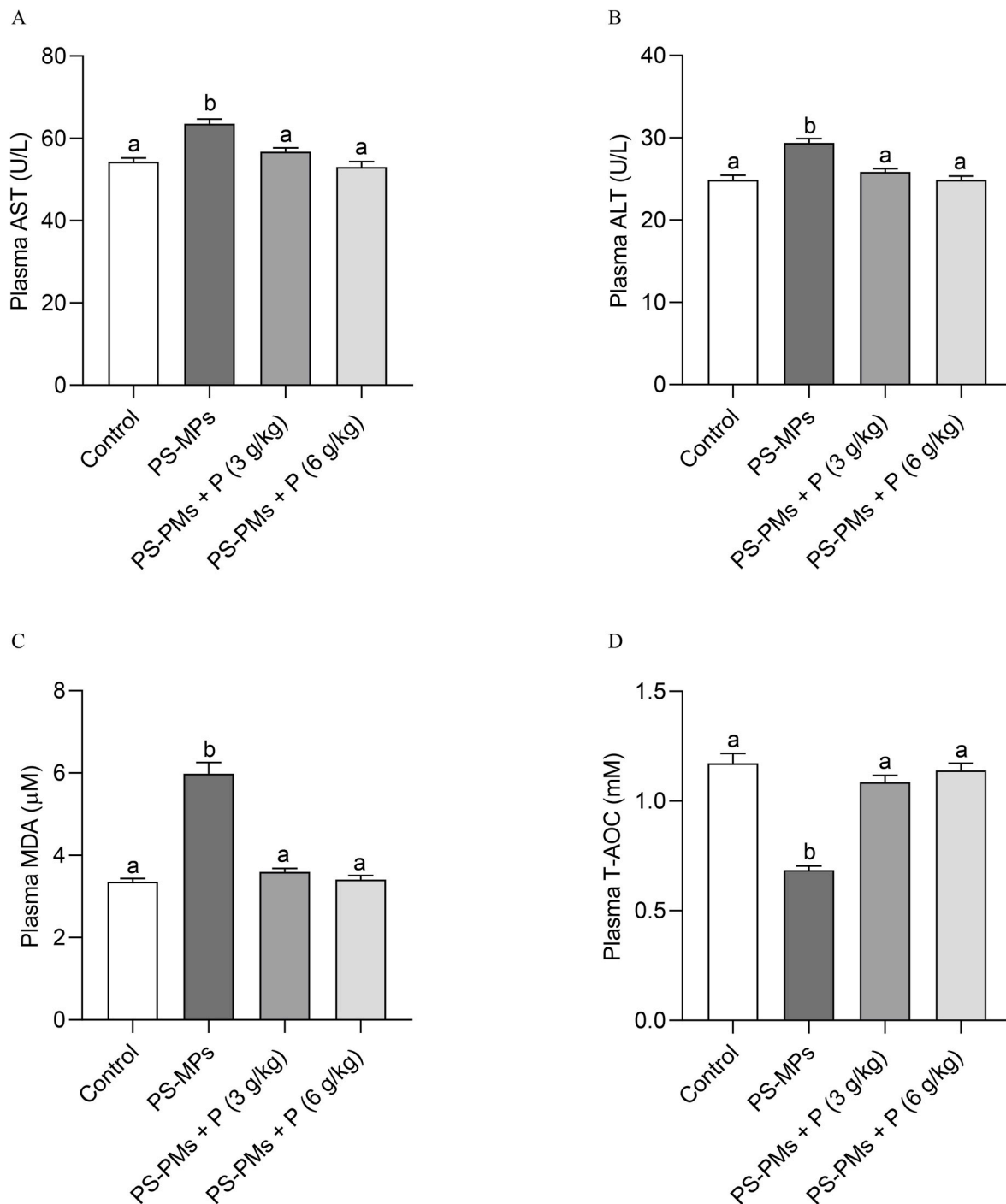


Fig. 3. Effect of probiotics on serum antioxidative status in PS-MPs-treated tilapia. (A) AST activity, (B) ALT activity, (C) MDA concentrations, and (D) T-AOC. Data are shown as mean $\pm$ SEM ($n = 12$ Control, 16 PS-MPs, 10 PS-MPs + P (3 g/kg), 12 PS-MPs + P (6 g/kg)). Different superscript letters in each column show significant differences ($p < 0.05$, one-way ANOVA).

antioxidant parameters GSH and GSSG in experimental fish were shown in Fig. 4F and G, respectively. PS-MPs exposure induced a significant decrease in GSH levels accompanied by increases in GSSG levels in the liver. Conversely, the augmentation of GSSG was prominently suppressed without affecting GSH when fish were pre-fed with probiotics at doses of 3 and 6 g/kg. Thus, liver redox status (as indicated by GSH/GSSG ratio) was restored in probiotics-supplemented fish (Fig. 4H).

3.5. Effects of probiotics on the transcription of selected cytokines following PS-MPs exposure in liver

To test whether probiotics mitigate the effect of PS-MPs exposure on inflammation, we evaluated the transcription of selected genes related to the inflammatory markers (CRP, TNF α , and IL-1 β). The transcriptions of *Crp* and *Tnfa* were significantly upregulated after PS-MPs exposure, but PS-MPs induction of *Crp* and *Tnfa* mRNA expression could be blocked by probiotics administration (Fig. 5A and B). In parallel experiments, there was no significant difference in *Il-1 β* after PS-MPs

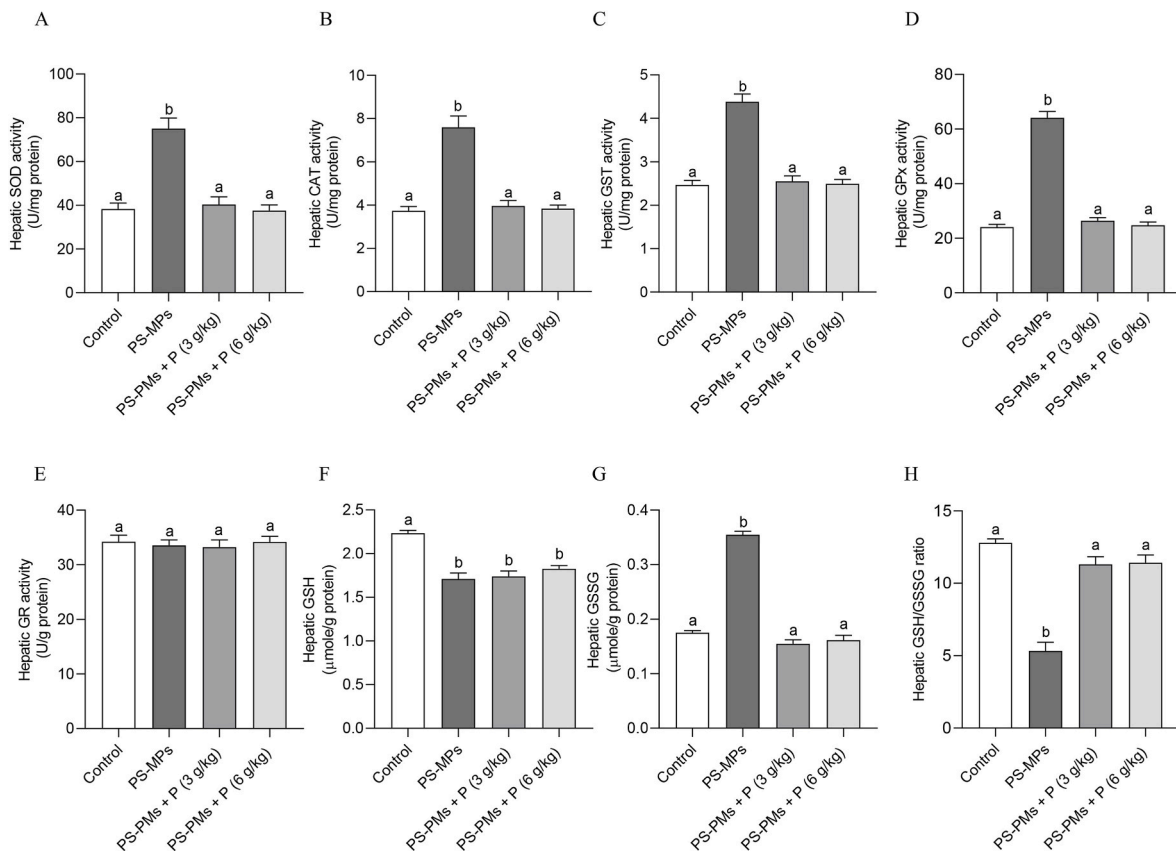


Fig. 4. Effect of probiotics on antioxidant enzyme activities and non-enzymatic antioxidant levels in the liver of tilapia treated with PS-MPs. (A) SOD activity, (B) CAT activity, (C) GST activity, (D) GPx activity, (E) GR activity, (F) GSH concentration, (G) GSSG concentration, and (H) GSH/GSSG ratio. Data are shown as mean $\pm$ SEM ($n = 12, 16, 10$, and 12 for above-mentioned four groups). Different superscript letters in each column show significant differences ($p < 0.05$, one-way ANOVA).

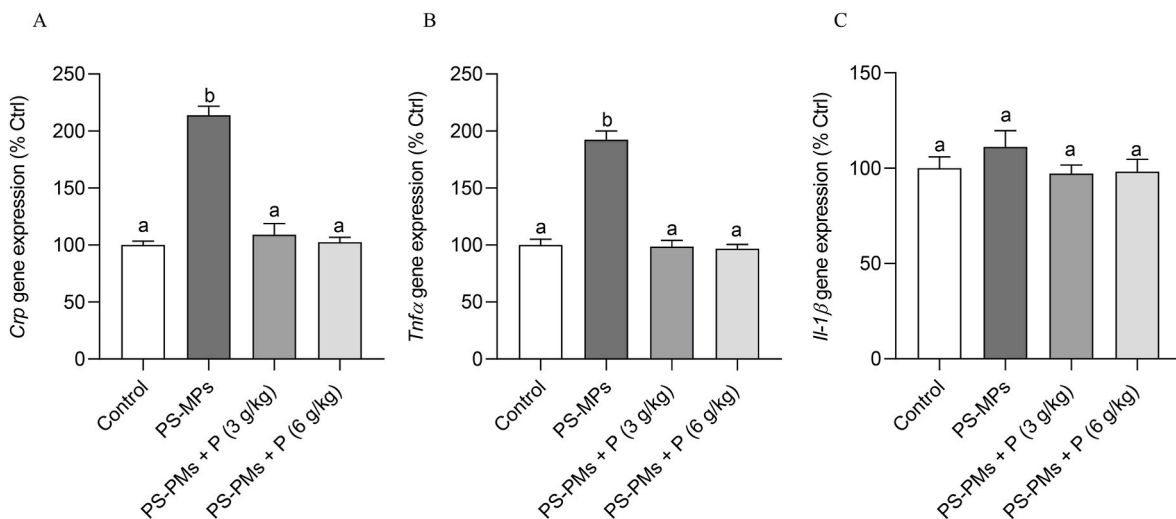


Fig. 5. Effect of probiotics on the hepatic gene expression of *Crp* (A), *Tnfa* (B), and *Il-1 β* (C) under PS-MPs treatment. Data are shown as mean $\pm$ SEM ($n = 12, 16, 10$, and 12 for above-mentioned four groups). Different superscript letters in each column show significant differences ($p < 0.05$, one-way ANOVA).

exposure in the presence or absence of probiotics treatments (Fig. 5C).

3.6. Probiotics prevented PS-MPs induction of ROS levels and MAPKs signalings

To determine whether probiotics negated the effect of PS-MPs exposure on ROS production, pre-feeding probiotics or the ROS scavenger NAC were co-administered with PS-MPs. Exposed to $0.5 \mu\text{M}$ PS-

MPs exhibited a significant increase in hepatic ROS level, while PS-MPs-induced ROS production was prevented by pre-feeding with probiotics (6 g/kg) or NAC treatment (Fig. 6A). MAPKs are important regulators of diverse cellular processes and stress responses [49]. To investigate the role of MAPK pathways in probiotics-mediated protection against PS-MPs-induced oxidative stress, Western blotting was used to assess the level of expression of MAPK signaling proteins. The phosphorylation levels of p38 MAPK (Fig. 6B), ERK (Fig. 6C), and JNK

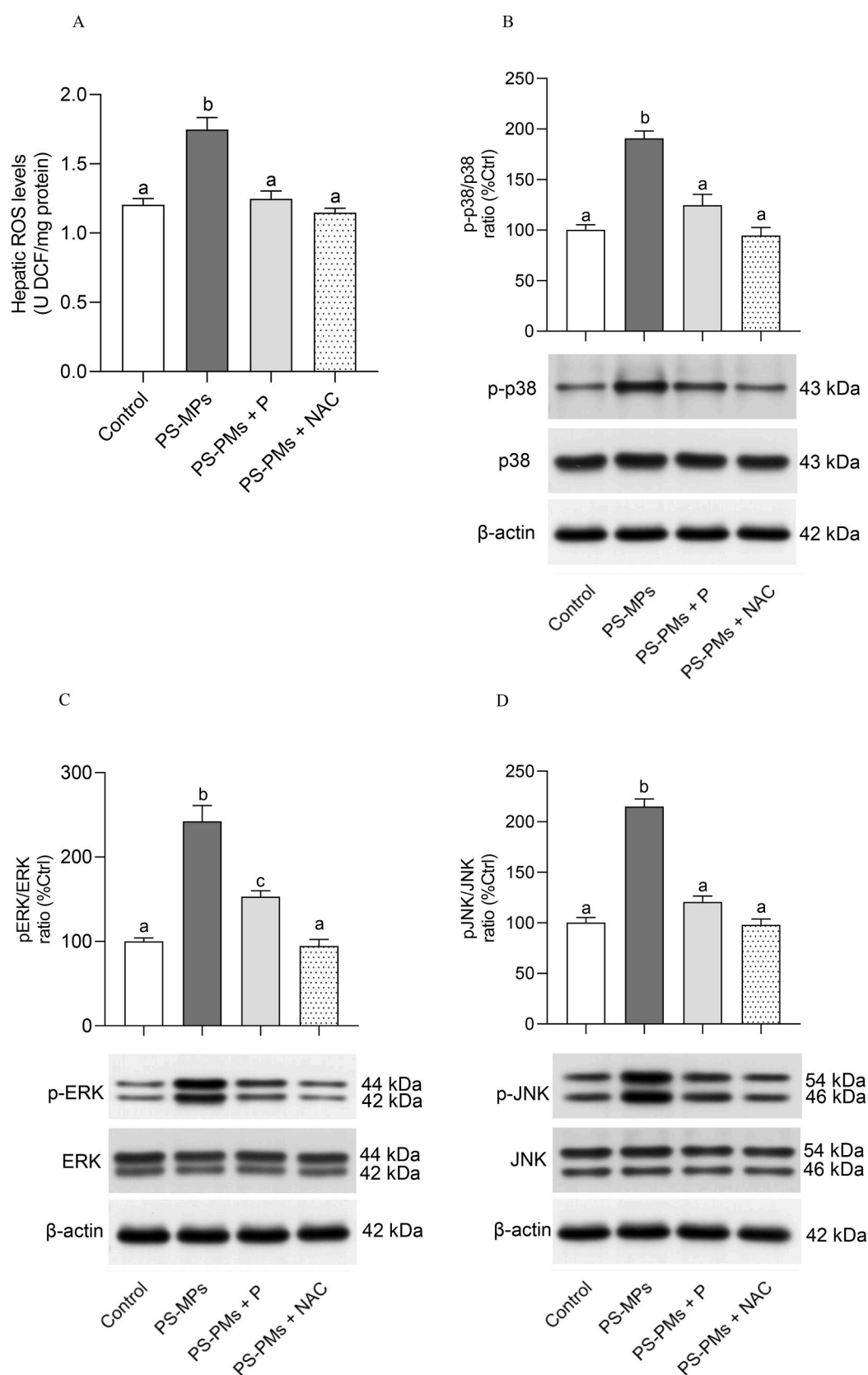


Fig. 6. Effect of probiotics or NAC on ROS production and phosphorylation of MAPK signaling proteins under PS-MPs exposure. The liver lysates were prepared from the following groups (1) Control, (2) 1 mg/L PS-MPs, (3) 1 mg/L PS-MPs + pre-feeding probiotics (6 g/kg), and (4) 1 mg/L PS + 100 mg/kg/d NAC. (A) ROS production. Data are shown as mean $\pm$ SEM with liver extracts from 6 to 9 fish per group. Densitometric quantitation of the phosphorylated forms of (B) p38, (C) ERK, and (D) JNK. The ratio of phosphorylated/total form of the proteins of interest was used as a measure of the activation of each signal transduction protein. The protein intensity is averaged from three blots and represented as mean $\pm$ SEM. Different superscript letters in each column show significant differences ($p < 0.05$, one-way ANOVA).

(Fig. 6D) were markedly increased in PS-MPs-exposed liver tissues compared with the control group, indicating that activation of MAPK occurs when PS-MPs induce hepatic oxidative stress. In contrast, pretreatment with 6 g/kg of probiotics or co-administered with NAC partially or fully prevented MAPK phosphorylation without significant changes in total proteins of p38 MAPK, ERK, and JNK.

4. Discussion

Probiotic bacteria have been shown both *in vivo* and *in vitro* to possess significant antioxidant capacities [19]. However, little information is available about the effects of probiotics on MPs-induced oxidative stress. The present study demonstrated that tilapia fed a commercial diet supplemented with a multi-species probiotics preparation alleviated PS-MPs-induced hepatic oxidative stress. To our knowledge, this is the first report regarding the effect of probiotics on MPs-induced oxidative stress in vertebrates.

At present, the study linking MPs to gut microbiome disruption in fish is still controversial. In European sea bass (*Dicentrarchus labrax*), a reduction in *Bacillus* and LAB could be observed after 60 days of exposure to PP-MPs [50], whereas neither dysbiosis nor altered bacterial diversity was observed after exposure to MPs in cultured European sea bass [51]. As a result of our study, although there was no significant impact on the total viable counts after PS-MPs exposure, LAB and *Bacillus* spp. significantly decreased when compared to untreated control fish. Furthermore, probiotic feeding resulted in higher LAB, *Bacillus* spp., and *enterococci* counts in the gastrointestinal tract of tilapia. This suggests that AquaStar® Growout can modulate the gastrointestinal microbiome and may have a stabilizing effect on the community. It also needs to be emphasized that manual counting of viable bacteria colonies on agar plates is time-consuming. Thus, molecular-based analysis in the future study should be conducted to identify which intestinal bacteria species are affected and how these changes relate to tilapia health.

The liver is highly adaptive to oxidative damage and is the main organ used to study the effects of pollutants in fish [52]. Although the research linking MPs with fish liver toxicity is still limited, the histopathological analysis showed that both 5 µm and 70 nm PS-MPs caused oxidative stress in the zebrafish (*Danio rerio*) liver after 7 days of exposure [53]. In addition, PE-MPs of 2 µm and 10 µm cause mild fibrosis in marine medaka (*Oryzias latipes*) liver tissues following 60 days of exposure [54]. In line with these findings, the present study provided evidence that exposure to 0.5 µm PS-MPs for two weeks resulted in vacuolation in hepatocytes, degeneration of the nuclei, and erythrocyte infiltration into blood sinusoids. These changes could be attributed to the direct toxic effects of PS-MPs on hepatocytes. ALT and AST are the most important indexes that reflect liver function. In our current studies, the 0.5 µm PS-MPs exposures group showed higher serum AST and ALT levels, reflective of hepatocellular injury due to oxidative stress. Consistently, PS-MPs significantly increased AST and ALT enzyme levels in common carp (*Cyprinus carpio*) [55] and tilapia [34]. However, after probiotic treatments, liver histological changes, as well as increases in serum marker enzymes (AST and ALT) were markedly attenuated, suggesting that AquaStar® Growout was effective in alleviating PS-MPs-induced hepatic oxidative stress in our present study. As far as we know, MDA is the final lipid peroxidation product and is frequently used to define oxidative stress [56]. In this study, PS-MPs could increase serum MDA levels, indicating the occurrence of PS-MP-induced oxidative stress and lipid peroxidation. Our present findings were also consistent with previous reports on tilapia [31], yellow croaker (*Larimichthys crocea*) [57], and European perch (*Perca fluviatilis*) [58], where PS-MPs exposure caused a significant increase in MDA levels. However, a few contrary studies also reported that the levels of MDA did not observe a significant increase after MPs exposure in the liver of European sea bass [59], tilapia [7], the common goby (*Pomatoschistus microps*) [60], and zebrafish [61]. These apparently contradictory results might be related to the different sizes and types of

the MPs and the different exposure methods used in these studies. T-AOC is reduced in conditions associated with oxidative stress. In agreement with previous findings in tilapia [31], the PS-MP exposure decreased T-AOC levels. Nevertheless, the elevation of MDA and the decline of T-AOC in serum were significantly restored as a result of pretreatment with AquaStar® Growout (3 and 6 g/kg).

Then we further explored the antioxidant mechanisms of probiotics behind liver injury caused by PS-MPs. The enzymes SOD and CAT provide primary antioxidant protection against free radical oxidative damage [62]. Our studies found an increase in SOD activity in the liver, suggesting that after MP exposure, the excess superoxide radical needs to be converted into H₂O₂ to prevent cellular oxidative stress. In parallel studies, the increase in CAT activities in the liver might be in response to H₂O₂ produced by SOD activity since CAT is responsible for the detoxification of H₂O₂ to oxygen and water. These findings were accordant to the previous reports in tilapia [31] and zebrafish [53], where hepatic SOD and CAT activities were elevated after PS-MPs exposure. On the contrary, zebrafish liver SOD and CAT activities did not show significant differences after exposure to MP for 7 days [63], suggesting that the liver requires a minimum exposure term to trigger the antioxidant system. It is noteworthy, however, that pre-feeding probiotics (3 and 6 g/kg) reversed the upregulation of SOD and CAT mediated by PS-MPs, suggesting that probiotics have antioxidative properties to regulate SOD and CAT activities for alleviating oxidative stress in the liver of tilapia.

GST, GPx, GR, GSH, and GSSG comprise the GSH-related antioxidant system, which constitutes an important endogenous antioxidant system [64]. The essential role of GST is to catalyze the conjugation of GSH with a wide variety of compounds, which result in the formation of the corresponding GSH-conjugates and subsequently facilitate their clearance from the body [65]. We found that exposure to PS-MPs induced GST activity, which was consistent with previous findings in gilthead seabream (*Sparus aurata*) [66] and zebrafish exposed to PE-MPs [67], as well as in goldfish (*Carassius auratus*) exposed to PVC-MPs [68]. GPx uses GSH to degrade H₂O₂ into H₂O and converts GSH into GSSG, whereas GR reduces GSSG to GSH in its sulfhydryl form. As previously described, GPx activity was elevated in the liver of juvenile yellow croaker exposed to PS-MPs [57] and in gilthead seabream after PVC-MPs exposure [69]. Conversely, GPx activity was decreased in the livers of common carp larvae exposed to the PVC-MPs [70]. Here, the increased GPx activity was not accompanied by any changes in GR activities in the PS-MPs exposed group. The possible explanation is that the GPx showed effective antioxidant activity by converting GSH to GSSH, which disposed of oxidative stress. This assumption is consistent with our present data, showing the decline in GSH and concomitant increase in GSSG concentrations in the PS-MPs exposed liver. In line with our findings, the content of GSH decreased significantly after exposure to 100 and 1000 µg/L 5 µm and 50 µm PS-MPs in larval zebrafish [71]. Similarly, a reduction in GSH level but an increase of GSSG have also been found in the guppy (*Poecilia reticulata*) after PS-MPs treatment for 28 days [72]. On the contrary, GSH activities in PS-MPs treatments significantly increased in the liver of the discus fish (*Symphysodon aequifasciatus*) [73]. Changes in GSH antioxidants in different fish suggested a ductile response of fish in reply to MPs-induced oxidative stress. However, the disruption of GST and GPx activities triggered by PS-MP exposure could be restored to control levels by pre-fed probiotics. Furthermore, the protective effects of probiotics on PS-MPs-treated tilapia were associated with significant decreases in oxidized glutathione (GSSG) and concomitant restoration of the cellular redox status (as indicated by GSH/GSSG ratio) to levels found in fish not treated with PS-MPs. Together, the inactivation of GSH-related antioxidant systems other than GR and GSH could be associated with the protective effect of probiotics against PS-MPs-mediated hepatic oxidative stress in tilapia.

The toxicity of MPs in cells is primarily caused by oxidative stress, which produces ROS, and these radicals then cause inflammation [74]. CRP is produced by the liver as a biomarker for active and chronic inflammation [75]. The pro-inflammatory cytokines (TNFα and IL-1β)

mediate the innate immune response, and dysregulation of the innate immune response contributes to pathogenesis [76]. In our study, the gene expressions of *Crp* and *Tnfa* were significantly increased with the presence of PS-MPs, suggesting that PS-MPs induced inflammation in tilapia. Similar to our findings, the expression of *Tnfa* was significantly increased with the presence of MPs in yellow catfish (*Pelteobagrus fulvidraco*) [77] and zebrafish [78]. In contrast, the relative expression of *Il-1 β* was not influenced by the PS-MPs, which is consistent with previous reports on European sea bass and gilthead seabream [79]. Of note, probiotics supplementation could block PS-MPs-induced *Crp* and *Tnfa* gene expression, suggesting that probiotics could prevent the deleterious effects of PS-MPs on inflammation.

The activation of MAPKs by ROS has physiological effects on the cellular tissue [49]. In our study, ROS was significantly induced in response to 0.5 μ m PS-MPs, while PS-MPs-induced ROS levels were not observed with co-administration of probiotics or a ROS scavenger NAC. These observations confirm that probiotics mitigated PS-MPs-mediated hepatic toxicity by alleviating oxidative stress. Previous studies had shown that the PS-MPs exposure caused oxidative stress via JNK and p38 MAPK in Monogonont Rotifer [14] and *Daphnia magna* [17]. In the marine copepod, ERK and p38 MAPK were activated in response to MPs-induced oxidative stresses, while the p38 MAPK pathway plays a negative role [16]. Our data agreed with these studies, demonstrating that PS-MPs exposure can activate the p-p38 MAPK, p-JNK, and p-ERK. However, with co-administration of the antioxidant NAC, activation of p-p38, p-ERK, and p-JNK were not observed. A similar phenomenon was observed by pretreatment with the probiotics, suggesting that the blocking MAPK activities in tilapia involved the defense mechanism of probiotics against PS-MPs-induced hepatic oxidative stress.

In summary, the current study demonstrated that continuous supplementation with AquaStar® Growout ameliorated liver damage induced by PS-MPs, which exhibited hepatoprotective effects by reducing serum AST, ALT, and MDA levels while increasing T-AOC levels. In addition, administration of the probiotics prevented PS-MPs-induced oxidative effects not only through modulating the activity of antioxidant components (e.g., SOD, CAT, GPx, GST, and GSSG) and inflammation marks (e.g., *Crp* and *Tnfa*), but also through interfering with MAPK signaling in the liver of tilapia. In light of these observations, this study indicated that probiotics could serve as functional foods and therapeutic agents for preventing oxidative stress induced by PS-MPs in fish.

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CRediT authorship contribution statement

Rui Dong: Conceptualization, Data curation, Investigation, Methodology, Writing – review & editing. **Changlei Zhou:** Conceptualization, Methodology. **Shuyue Wang:** Conceptualization, Methodology. **Yisha Yan:** Conceptualization, Investigation. **Quan Jiang:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Project administration, Writing – review & editing.

Declaration of competing interest

No conflicts of interest, financial or otherwise, are declared by the authors.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fsi.2022.09.022>.

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